Suhaas Behara

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EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science in Mechanical Engineering - GPA: 3.80/4.00 Expected May 2028

- Dean's List Recipient | Fall 2025 and Spring 2026
- Semester Honors | Fall 2025 and Spring 2026

SKILLS

Software: Siemens NX, Fusion 360, SolidWorks, SolidWorks FEA, SolidWorks PDM, ANSYS Mechanical, KiCad, LTspice
Programming: Python, MATLAB, C++, Arduino, Git
Engineering: Rapid Prototyping, GD&T, 3D Printing, Additive Manufacturing, Machining

PROFESSIONAL EXPERIENCE

Operations Intern Manchester, NH
Riverstone Resources June 2026 - Aug 2026

- Automated recurring operational workflows using Microsoft Copilot and AI-driven tools, reducing manual spreadsheet analysis and reporting time by 60%+ while improving data consistency and accuracy
- Identified opportunity to enhance data efficiency and developed AI-assisted tools, detected missing and inconsistent information, and generated prioritized follow-up actions, saving 8+ hours per week of repetitive manual work
- Evaluated business processes and implemented AI-based automation solutions that streamlined document review, standardized reporting, and improved decision-making across the Ceded Reinsurance team

Undergraduate Research Assistant West Lafayette, IN
Purdue Mechanisms and Robotic Systems Lab (MARS) Aug 2026 – Present

- Designed compact mechanical components in Siemens NX for a vision-based photoelastic tactile sensor, balancing manufacturability, optical alignment, and integration with embedded sensing hardware
- Fabricated and iterated sensor prototypes to resolve packaging and assembly constraints, supporting force measurement and tissue characterization for robot-assisted minimally invasive surgery
- Validated component designs through prototype assembly and mechanical testing with doctoral researchers, translating test findings into design revisions for improved sensor performance and reliability

Project Manager West Lafayette, IN
EPICS MOBI Team Aug 2025 – May 2026

- Led a team of 10 engineers through two 16-week design cycles, coordinating tasks, timelines, and design reviews while driving decisions on manufacturing methods and system integration
- Engineered and optimized a protective casing using CAD and 3D printing, compressing advanced electronics into a softball-sized form factor that withstood 100+ impact tests while maintaining rechargeability and structural integrity
- Collaborated cross-functionally with the electronics team to integrate a 1,400 Hz, 80 dB system, balancing acoustic performance with mechanical constraints, and led rapid prototyping iterations to improve strength, weight efficiency, and manufacturability

PROJECTS

Active Orthosis West Lafayette, IN
ASME (Biomechanics) Jan 2026 – May 2026

- Designed and manufactured components for an active upper limb orthosis in Fusion 360 through iterative CAD modeling and dimensional tolerancing, improving manufacturability and range of motion
- Implemented embedded electronics by designing KiCad schematics and PCB layouts, selecting sensors, motor drivers, and power circuitry for reliable actuator control
- Validated rehabilitation prototype performance by integrating mechanical hardware with embedded control systems and resolving mechanical and electrical compatibility issues through iterative testing

Robotic Arm Westford, MA
Personal Project June 2026 – Aug 2026

- Designed, 3D printed, and assembled a 7 DOF tendon-driven robotic hand in Fusion 360, applying dimensional tolerancing and iterative mechanical design to improve assembly, actuator integration, and independent finger articulation
- Developed a Python-based hand tracking pipeline that detected 21 hand landmarks, solved inverse kinematics, and translated real-time finger motion into servo commands for intuitive robotic hand teleoperation
- Integrated embedded electronics, Arduino-based motor control, and electromechanical hardware into a functional prototype, validating motion through iterative calibration, testing, and mechanical refinement